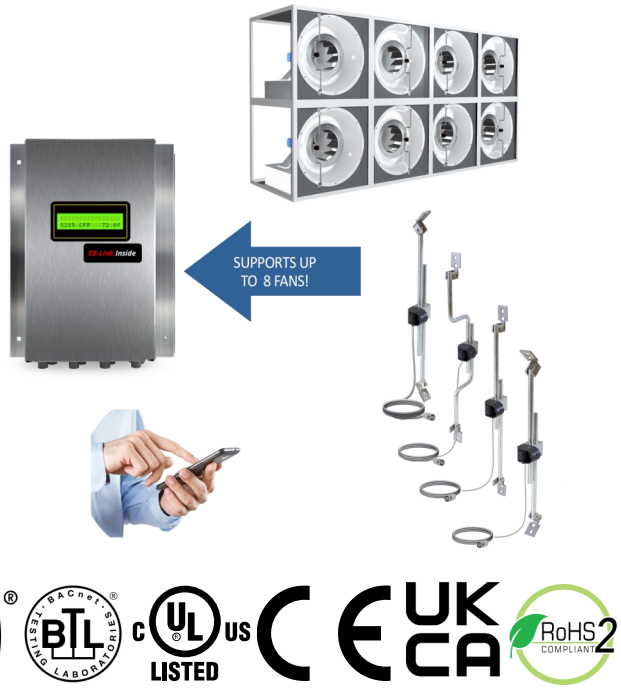
FAN AIRFLOW MEASUREMENT WITH TEMPERATURE AND ALARM CAPABILITY



PATENTS

- US Patent Nos.: 12,066,199; 12,066,205
- CA Patent Nos.: 3,069,531; 3,169,641
- EP Patent No.: 4081741
- MX Patent Nos.: 417881; 428776

TYPICAL APPLICATIONS

- Fan airflow tracking
- Air change verification and monitoring
- Individual fan performance monitoring and fault detection
- Air change verification and control

PRODUCT HIGHLIGHTS

- *"Plug and Play"* operation
- EBTRON exclusive bead-in-glass thermistor sensors
- Sensor nodes are individually calibrated at 16 airflow rates to NIST traceable standards
- 0 to 10,000 FPM calibrated range with percent-of-reading accuracy
- Actual (CFM) or mass (SCFM) airflow measurement
- Velocity-weighted temperature measurement between -20 °F to 160 °F
- Smart *Sensor Detection System* (*SDS*) continuously monitors for sensor and transmitter faults
- Standard FEP plenum rated cable between sensor probes and transmitter
- No compromise construction uses gold plated interconnects
- Unsurpassed connectivity options
- Four mounting styles
- *EB-Link* BLE interface to phone or tablet provides real-time monitoring and diagnostics
- Three-year warranty
- Toll-free customer support for the lifetime of the product

EBTRON ADVANCED THERMAL DISPERSION TECHNOLOGY

EBTRON pioneered bead-in-glass thermistor based thermal dispersion over 40 years ago. EBTRON's thermal dispersion technology relates the power dissipated by a self-heated thermistor to the airflow rate at one or more sensor nodes in an airstream. All EBTRON airflow monitoring systems use this time-tested thermal dispersion technology.

MODEL DESCRIPTION

The GTx108e-F/SI and GTx108e-F/DI are EBTRON's solution for accurate and repeatable airflow measurement in SWSI and DWDI fans. The GTx108e-F/An is EBTRON's solution for accurate and repeatable airflow measurement in fan arrays. One to eight fans are supported. Airflow, temperature and/or airflow alarming are available on all models. The GTx108e-F/An provides individual fan airflow rates and fan alarming with combination analog output/network models. Does not affect fan performance.

EBTRON, Inc., 1663 HWY 701 South, Loris SC 29569
800-2-EBTRON (1-800-232-8766)
EBTRON.com
sales@ebtron.com

GTx108e-F TECHNICAL SPECIFICATIONS

General

Probe and Sensor Node Configurations

Fan Arrays (less than or equal to 4 fans): 2 probes x 1 sensor node per probe or 1 probe x 1 sensor node per probe in each fan

Fan Arrays (greater than 4 fans): 1 probe x 1 sensor node per probe in each fan (8 probe maximum)

SWSI and DWDI fans: 2 probes x 1 sensor node per probe in each fan inlet

Installed Airflow Accuracy[1]

±(3% to 10%) of reading, depending on fan type and installation. May be improved by field adjustment using the Field Adjust Wizard (FAW) to a reliable reference

Sensor Node Averaging Method

Airflow: Independent, arithmetic average per fan

Temperature: Independent, velocity weighted average

Listings & Compliance

UL: 60730-1; CAN/CSA-E60730-1

CE: Yes (/NR option only)

UKCA: Yes (/NR option only)

BACnet International: BTL Listed (GTB108e, GTC108e, GTM108e and GTS108e transmitters)

FCC: This device complies with Part 15 of the FCC rules

RoHS: This device is RoHS2 compliant

Environmental Limits

Temperature:

Probes: -20 to 160 ^{0}F [-28.9 to 71.1 ^{0}C]

Transmitter: -20 to 120 ^{0}F [-28.9 to 48.9 ^{0}C]

Humidity: (non-condensing)

Probes: 0 to 100%

Transmitter: 5 to 95%

Individual Sensing Nodes

Sensing Node Sensors

Self-heated sensor: Precision, hermetically sealed, bead-in-glass thermistor probe

Temperature sensor: Precision, hermetically sealed, bead-in-glass thermistor probe

Sensing Node Housing

Material: Glass-filled Polypropylene

Sensor Potting Materials: Waterproof marine epoxy

Airflow Measurement

Accuracy: ±2% of reading to NIST-traceable airflow standards (includes transmitter uncertainty)

Calibrated Range: 0 to 10,000 fpm [0 to 50.8 m/s]

Calibration Points: 16

Temperature Measurement

Type: Velocity-weighted average

Accuracy: ±0.15^{0}F [0.08 ^{0}C] to NIST-traceable temperature standards (includes transmitter uncertainty)

Calibrated Range: -20 to 160 ^{0}F [-28.9 to 71.1 ^{0}C]

Calibration Points: 3

Sensor Probe Assembly

Mounting Rods

Material: Zinc plated steel

Mounting Brackets (Throat, Forward, Face, Flare)

Material: 304 stainless steel

Mounting Options & Size Limits

Throat: 6 to 66 inches [152.4 to 1676.4 mm] (throat diameter)

Face: 11 to 77 inches [279.4 to 1955.8 mm] (diameter at inlet entrance)

Forward: 6 to 64 inches [152.4 to 1625.6 mm] (diameter at inlet entrance)

Flare: 6 to 57 inches [152.4 to 1447.8 mm] (opening size at backdraft damper inlet)

Probe to Transmitter Cables

Type: FEP jacket, plenum rated CMP/FT6/CL2P, UL/cUL listed, -67 to 302 ^{0}F [-55 to 150 ^{0}C], UV tolerant

Standard Lengths: 10, 25, and 50 ft. [3.1, 7.6, and 15.2 m]

Connecting Plug: 9/16" [14.29 mm] nominal diameter

Transmitter

Power Requirement: 24 VAC (22.8 to 26.4 under load) @16V-A

PCB Connections: Gold-plated PCB interconnects, PCB edge fingers, and test points

User Interface: 2 line x16-character backlit LCD display and 4 button interface

B.A.S. Connectivity Options

All Transmitters: Three field selectable (0-5/0-10 VDC or 4-20mA), scalable and isolated analog output signals (AO1=airflow, AO2=temperature or alarm, AO3=Not Used).

GTA108e Transmitter: No additional connectivity to B.A.S.

GTC108e Transmitter[2]**:** One additional field selectable (BACnet MS/TP or Modbus RTU) and isolated RS-485 network connection

GTB108e Transmitter[2]**:** One additional isolated RS-485 and one additional isolated Ethernet network connections field configurable as follows:

- Single RS-485 connection to B.A.S. with field selectable BACnet MS/TP or Modbus RTU, and
- Single Ethernet connection to B.A.S. with simultaneously supported BACnet IP or BACnet Ethernet, Modbus TCP, and TCP/IP, or
- Single Ethernet connection to *EB-Bus* device(s)

GTM108e Transmitter[2]**:** One additional isolated Ethernet network connection field configurable as follows:

- Single connection to B.A.S. with simultaneously supported BACnet IP or BACnet Ethernet, Modbus TCP, and TCP/IP, or
- Single connection to *EB-Bus* device(s)

GTS108e Transmitter[2]**:** Two additional isolated Ethernet network connections field configurable as follows:

- Single connection to B.A.S. and single connection to *EB-Bus* device(s), or
- Daisy-chain connection to B.A.S., or
- Daisy-chain connection to *EB-Bus* device(s)

Note: B.A.S. network connections can simultaneously support BACnet IP or BACnet Ethernet, Modbus TCP, and TCP/IP.

GTU108e Transmitter: One additional USB connection for thumb drive data-logging of sensor node airflow rates and temperatures

Alarms

Airflow: Low and/or high user defined setpoint alarm

Fan (An models Type): Minimum airflow, % deviation from median airflow, or % deviation from maximum airflow stored in memory

System Status: Sensor diagnostic system trouble indication

***EB-Link* Bluetooth® low energy Interface for Android® and iPhone®:** Download individual sensor node airflow/temperature data, settings and diagnostics[3].

[1] Installed airflow accuracy is the actual system accuracy expected and includes sampling uncertainty of the sensor probes.

[2] Individual sensor node airflow rates and temperatures are available via the network.

[3] Order with the /NR option when RF devices are not permitted.

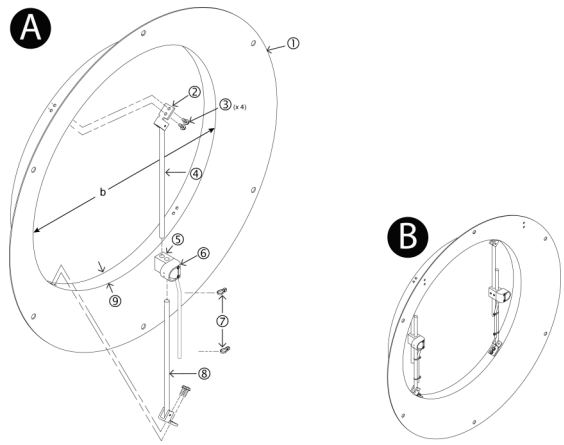
SENSOR PROBE INSTALLATION

Detailed installation guidelines for probes are provided in the O&M Manual for the following mounting styles.

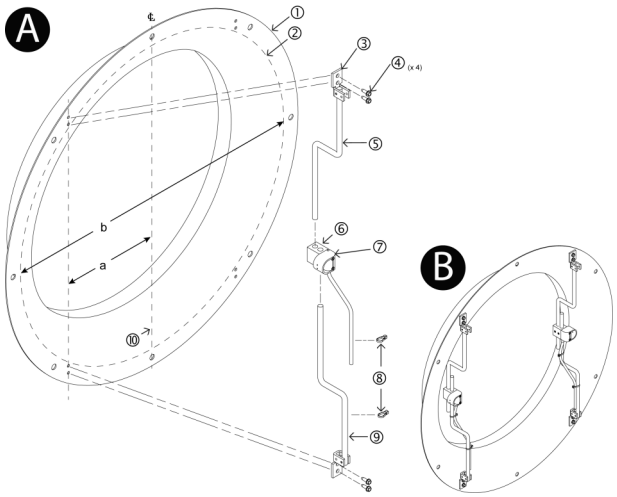
⚠ *Install sensor probes at location (if provided) indicated on probe hang tag to optimize implementation of the EB-Link Reader and IAQ Enforcer® accessories.*

Sensor Probe Installation Diagrams

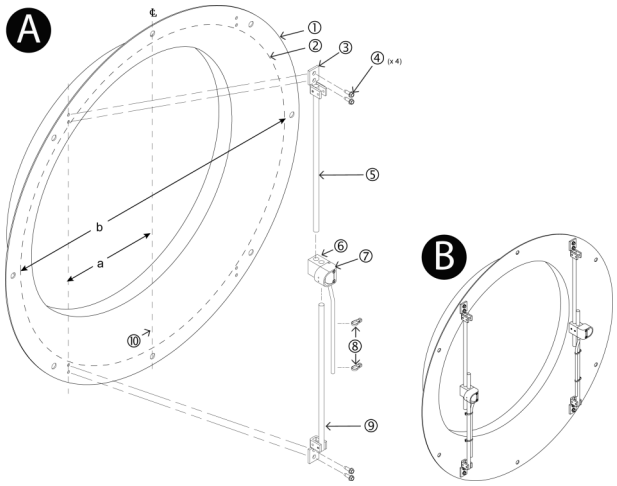
-F Probe Installation (Throat Mounting Rods)



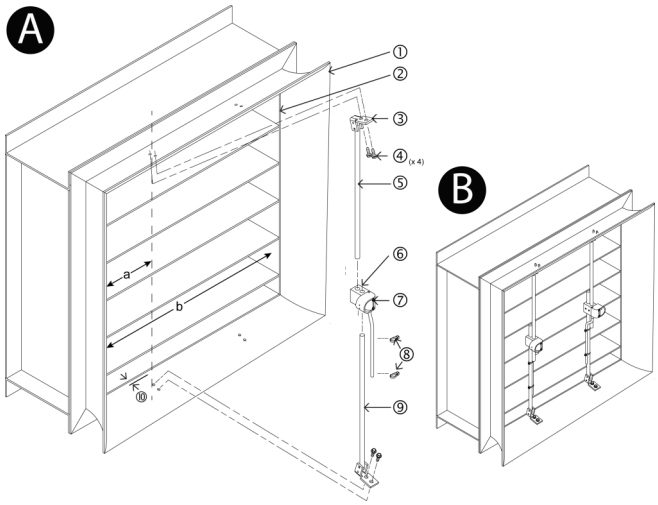
-F Probe Installation (Face Mounting Rods)



-F Probe Installation (Forward Mounting Rods)



-F Probe Installation (Flare Mounting Rods)



Inlet Probe Size and Mounting Style

	Inlet Size Range (Diameter at mounting location)						Damper Opening	
	Throat		Face		Forward		Flare	
Size Code	Min. (inches)	Max. (inches)	Min. (inches)	Max. (inches)	Min. (inches)	Max. (inches)	Min. (inches)	Max. (inches)
000	≥ 6	≤ 7	NA	NA	NA	NA	NA	NA
00	≥ 7	≤ 8	NA	NA	NA	NA	NA	NA
0	≥ 8	≤ 12	NA	NA	≥ 6	≤ 9	≥ 6	≤ 9
1	≥ 10	≤ 15	≥ 11	≤ 13	≥ 7	≤ 13	≥ 8	≤ 11
2	≥ 14	≤ 23	≥ 13	≤ 18	≥ 10	≤ 18	≥ 10	≤ 15
3	≥ 18	≤ 32	≥ 18	≤ 25	≥ 14	≤ 25	≥ 12	≤ 20
4	≥ 27	≤ 49	≥ 25	≤ 35	≥ 19	≤ 35	≥ 16	≤ 27
5	≥ 35	≤ 66	≥ 34	≤ 50	≥ 27	≤ 50	≥ 21	≤ 38
6	NA	NA	≥ 47	≤ 77	≥ 40	≤ 64	≥ 31	≤ 57

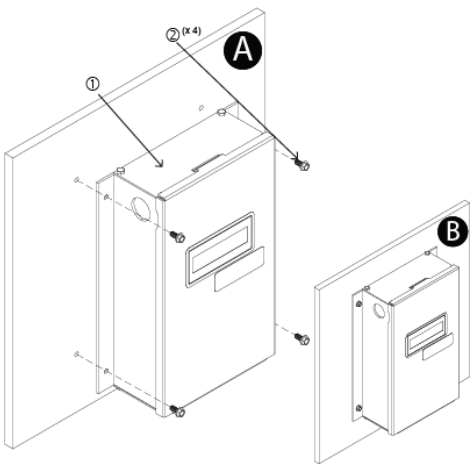
TRANSMITTER INSTALLATION AND WIRING

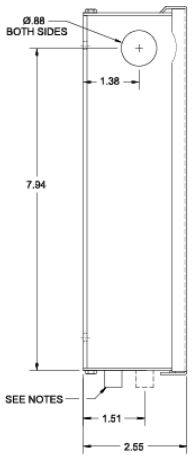
Detailed installation guides for transmitters are provided in the O&M Manual for the connectivity options that follow. All GTx108e transmitters are provided with the base analog output configuration (x = A). Additional connectivity may be obtained by an additional output card located below the display on the main circuit board.

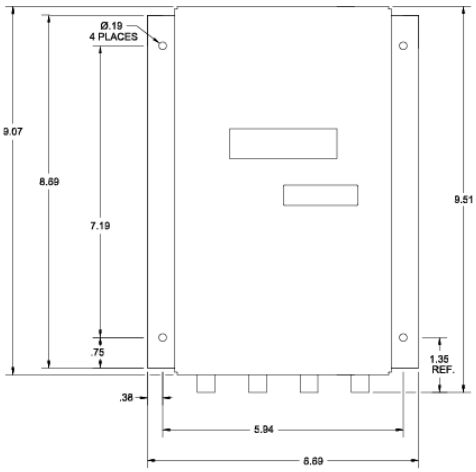
Locate the transmitter where all of the sensor probe connecting cables will reach the transmitter and it is protected from moisture, rain, and snow.

Transmitter Mounting Diagram and Dimensions

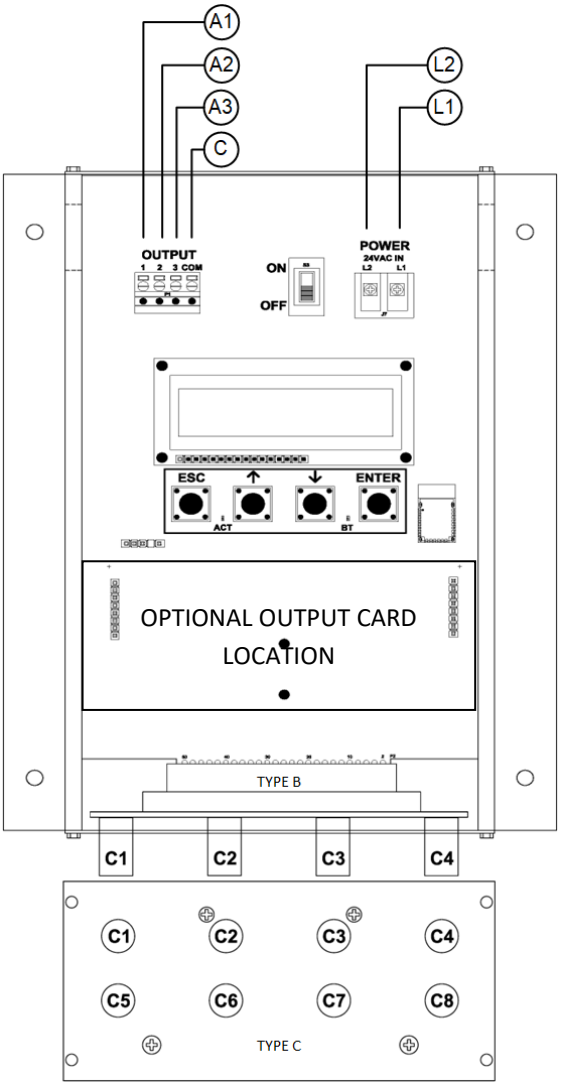
Advantage IV Gold Series Transmitter Installation

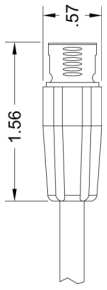






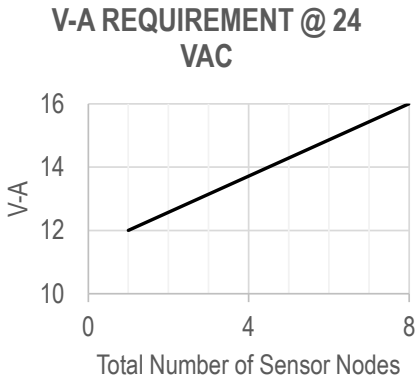
All Models - Power Requirement, Probe Connections, and Analog Output Connections





⚠ *Push in **keyed** connector plugs (sensor data is stored in serial memory chip in connector plug). **DO NOT TWIST!***

ⓘ *Cables have an FEP plenum rated jacket that are UV tolerant and suitable for operation over the entire operating temperature range of the device.*



TRANSMITTER CONNECTIONS

Power		Analog Out (isolated)				Type	
L1	L2	A1	A2	A3	C	B	C
24 VAC (hot)	24 VAC (neutral)	Airflow +	Temperature or Alarm +	N/A	Signal Common	4 probes x 1 sensors/probe	8 probes x 1 sensors/probe

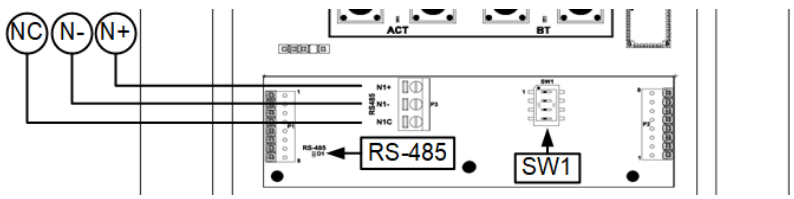
Connect each analog output signal required to the host B.A.S. using shielded twisted-pair wire. Properly terminate the shield (typically at the B.A.S.).

⚠ *If twisted pair wire and/or shielded cable is not used, extraneous electrical noise can be picked up between the transmitter and host control panel.*

ⓘ *Analog output signals are electrically isolated from the main circuit board and power supply.*

ⓘ *Transmitters are preconfigured for 4-20mA. Consult the Startup Guide for the appropriate model to change the output to either 0-5 VDC or 0-10 VDC.*

GTC108e-F - RS-485 Connections



RS-485 (isolated)		
N+	N-	NC
Network +	Network -	Network Common

GTC108e-F - RS-485 Connections (continued)

Use a 3-conductor network cable meeting the corresponding BACnet or Modbus standards. Ensure that all three connections, N+, N- and NC are connected.

ⓘ *Refer to the O&M Manual for information regarding network configuration, BACnet objects, and Modbus registers.*

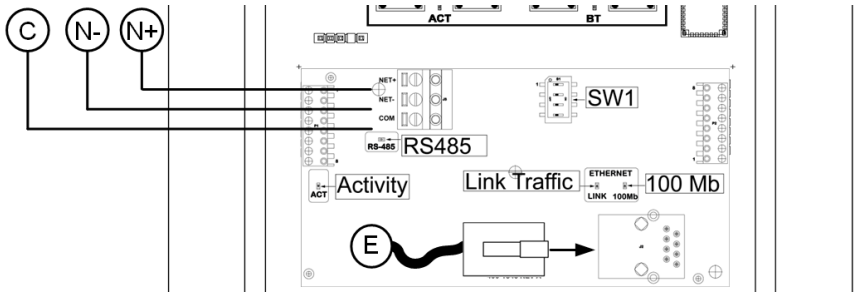
ⓘ *Transmitters are preconfigured for BACnet MS/TP. Consult the Startup Guide for the appropriate model to change the output to Modbus RTU network protocol.*

⚠ *If a 2-conductor network cable or other non-conforming cable is used, network speed, length and reliability may be compromised or network failure may occur.*

SW1—RS-485 TERMINATION DIP SWITCH POSITIONS

1	2	3	4	TERMINATION
OFF	OFF	OFF	OFF	No termination (default)
OFF	ON	ON	OFF	End of Line
ON	OFF	OFF	ON	Fail-Safe Bias

GTB108e-F - RS-485 and Ethernet Connections



RS-485 (isolated)			Ethernet
N+	N-	NC	E
Network +	Network -	Network Common	RJ-45 CAT5 or greater

If RS-485, use a 3-conductor network cable meeting the corresponding BACnet or Modbus standards. Ensure that all three connections, N+, N- and NC are connected.

ⓘ *Refer to the O&M Manual for information regarding network configuration, BACnet objects, and Modbus registers.*

ⓘ *Transmitters are preconfigured for BACnet MS/TP. Consult the Startup Guide for the appropriate model to change the output to Modbus RTU network protocol.*

⚠ *If a 2-conductor network cable or other non-conforming cable is used, network speed, length and reliability may be compromised or network failure may occur.*

SW1—RS-485 TERMINATION DIP SWITCH POSITIONS

1	2	3	4	TERMINATION
OFF	OFF	OFF	OFF	No termination (default)
OFF	ON	ON	OFF	End of Line
ON	OFF	OFF	ON	Fail-Safe Bias

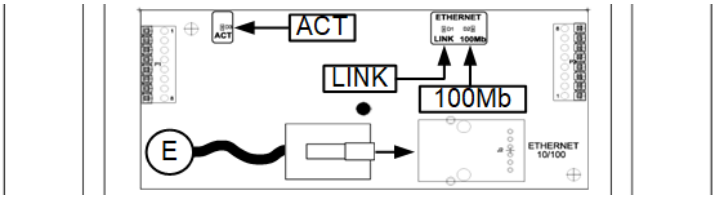
If Ethernet, use a standard RJ-45 Ethernet connection (T-568A or T-568B) and CAT-5 cable or higher.

ⓘ *Refer to the O&M Manual for information regarding network configuration, BACnet objects, and Modbus registers.*

ⓘ *BACnet Ethernet, BACnet IP, Modbus TCP or TCP/IP protocols are set during network configuration. The GTB108e-F supports simultaneous operation of all protocols.*

ⓘ *EB-Bus network configuration is set by EB-Bus Ethernet host. Refer to O&M Manual for information regarding EB-Bus network configuration.*

GTM108e-F - Ethernet Connection



Ethernet
E
RJ-45 CAT5 or greater

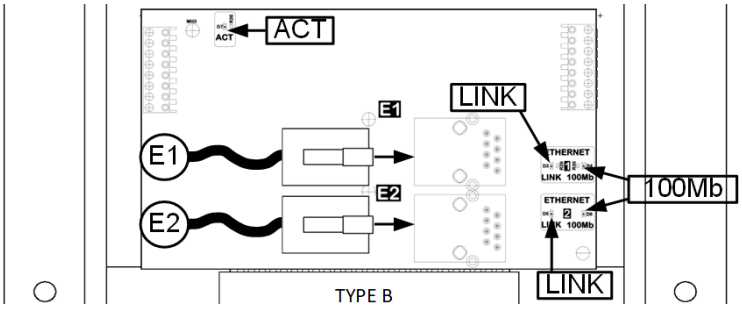
Use a standard RJ-45 Ethernet connection (T-568A or T-568B) and CAT-5 cable or higher.

ⓘ *Refer to the O&M Manual for information regarding network configuration, BACnet objects, and Modbus registers.*

ⓘ *BACnet Ethernet, BACnet IP, Modbus TCP or TCP/IP protocols are set during network configuration. The GTM108e-F supports simultaneous operation of all protocols.*

ⓘ *EB-Bus network configuration is set by EB-Bus Ethernet host. Refer to O&M Manual for information regarding EB-Bus network configuration.*

GTS108e-F - Dual Ethernet Connection



Ethernet	
E1	E2
RJ-45 CAT5 or greater	RJ-45 CAT5 or greater

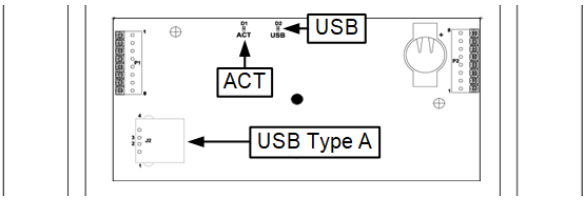
Use standard RJ-45 Ethernet connections (T-568A or T-568B) and CAT-5 cable or higher.

ⓘ *Refer to the O&M Manual for information regarding network configuration, BACnet objects, and Modbus registers.*

ⓘ *BACnet Ethernet, BACnet IP, Modbus TCP or TCP/IP protocols are set during network configuration. The GTS108e-F supports simultaneous operation of all protocols.*

ⓘ *EB-Bus network configuration is set by EB-Bus Ethernet host. Refer to O&M Manual for information regarding EB-Bus network configuration.*

GTU108e-F - USB Datalogger



Transmitters with a USB data logger log the average airflow, average temperature, individual sensor node airflow and the individual sensor node temperature at 5-minute intervals using Universal Time Coordinated (UTC) based on an onboard real-time clock, whenever power is applied to the transmitter. Data files are automatically appended on power-up.

Enable the USB WRITE parameter to start logging data.

Disable the USB WRITE parameter stop logging data.

⚠ *Always set the USB WRITE parameter to "OFF before removing the USB memory device to avoid data loss/and or damage.*

STARTUP

Detailed startup guides are provided in the O&M Manual.

VERIFICATION, TROUBLESHOOTING, AND MAINTENANCE

Verification and troubleshooting is discussed in detail in the O&M manual.

EBTRON airflow monitoring devices are factory calibrated and should not require periodic maintenance or recalibration in most HVAC environments. If the sensor probes are installed in a location that is subject to excessive dust, lint, or other airborne debris accumulation, filtration at the source or access for cleaning should be provided. Please note that this issue is not specific to EBTRON or EBTRON technology. Consult EBTRON to discuss specific concerns or suggested remedies if airborne particulates are a concern.

APPLICATION REVIEW REQUESTS

EBTRON and/or your local representative will review the application and make recommendations on the proper use of this device prior to requesting a quotation or placing an order. Provide as much of the following information as possible to facilitate the review process:

- √ Service: SA, RA, OA, EA, Other (specify)
- √ Duct or opening size where the probes will be installed
- √ Minimum and maximum expected airflow rates
- √ Minimum and maximum expected airstream temperatures
- √ Immediate up and downstream disturbance and the distance between disturbances
- √ Mechanical drawings or sketches
- √ Sequence of operation
- √ Control schematic

QUOTATIONS AND ORDERING

Use the model code below to request a quote or place an order. Quotations and orders are provided by your local EBTRON representative. Provide a location name up to sixteen characters to facilitate use of the EB-Link Reader phone/tablet application and IAQ Enforcer® accessories. The location name is displayed on the EB-Link Reader and IAQ Enforcer Smart Display Panel SDX-1000 in lieu of the transmitter serial number.

GT 108e-F / / :

Probe Cable Length
10 = 10 ft., 25 = 25 ft., 50 = 50 ft.
Note: Custom lengths between 3 ft. and 50 ft. are available

Inlet Probe Size Code
000, 00, 0, 1, 2, 3, 4, 5, 6

Mounting Style
1 = Throat, 2 = Face, 3 = Forward, 4 = Flare

Bluetooth Radio for EB-Link Reader
{leave blank} = Provided, NR = No radio

Application
SI = SWSI, DI = DWDI, An = Fan Array (n = number of sensor probes)

Connectivity
A = isolated analog outputs (AO1 = airflow, AO2 = temperature or alarm)
B = isolated analog outputs plus RS-485 field selectable BACnet MS/TP or Modbus RTU plus Ethernet field selectable BACnet Ethernet, BACnet IP, Modbus TCP, or *EB-Bus*
C = isolated analog outputs plus RS-485 field selectable BACnet MS/TP or Modbus RTU
M = isolated analog outputs plus Ethernet field selectable BACnet Ethernet, BACnet IP, Modbus TCP, or *EB-Bus*
S = isolated analog outputs plus dual Ethernet field selectable BACnet Ethernet, BACnet IP, Modbus TCP, or *EB-Bus*
U = isolated analog outputs plus USB thumb drive datalogger